3. Exploratory Data Analysis

Get to know your data

Univariate analysis

Definition

Univariate analysis means analyzing one variable at a time ("uni" = one).
It's the simplest form of data analysis, used to understand the distribution, central tendency, and spread of a single variable.

Type of Variable	Typical Techniques
Numerical	Mean, median, mode, variance, histogram, box plot
Categorical	Frequency counts, bar chart, pie chart

Univariate analysis

Measuring Central tendencies

Mean

There are 3 variations of the mean (average) used in univariate analysis to represent the center of a dataset in slightly different ways.

Arithmetic Mean: The mean is the most common measure of central tendency. It's simply the sum of all values divided by the number of values.

$$\text{Mean} = \frac{\sum x_i}{n}$$

n: the size of the population

If the exam scores are : 60, 70, 80, 90, 100
Then **Mean = (60+70+80+90+100) / 5 = 80**

Univariate analysis

Mean

There are 3 variations of the mean (average) used in univariate analysis to represent the center of a dataset in slightly different ways.

Trimmed mean: the trimmed mean is the mean calculated after removing a small percentage of the lowest and highest values.

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

n is sample size and N is population size.

Data = [10, 12, 13, 14, 15, 100]

10% trimming removes one smallest and one largest value → [12, 13, 14, 15]

Then **Mean = (12 + 13 + 14 + 15) ÷ 4 = 13.5**

Univariate analysis

Mean

There are 3 variations of the mean (average) used in univariate analysis to represent the center of a dataset in slightly different ways.

Weighted mean: assigns different importance (weights) to each value. Some observations contribute more to the average than others.

$$\overline{x} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$$

x_i : value, and w_i: weight of that value

Example: A student's grades: Homework (weight 20%): 80; Test (weight 30%): 70; Final EMD (weight 50%): 90

Weighted Mean=(0.2×80)+(0.3×70)+(0.5×90)/ (0.2+0.3+0.5) =82

Univariate analysis

Type of Mean	Used For	When to use	Example
Arithmetic Mean	Simple average	The data has no extreme values (outliers). All observations are equally important.	Average temperature
Trimmed Mean	Robust measure	You want a more robust mean (less influenced by outliers). Especially useful in economics or social data (e.g., income, prices).	Average income without extremes
Weighted Mean	When values have weights	Some data points are more significant than others (e.g., weighted grades, index numbers, survey sampling).	Final grade calculation

Univariate analysis

Median

The median is the middle value of a dataset when the values are arranged in order (from smallest to largest).

It divides the dataset into two equal halves:

- 50% of observations are below the median
- 50% are above the median

Useful for:

Works well with skewed data (like income or housing prices).
Not affected by extreme values.
Helps describe the “typical” case in non-symmetric distributions.

Univariate analysis

Median

The median is the middle value of a dataset when the values are arranged in order (from smallest to largest).

How to Calculate the Median

Step 1 : Order the data (ascending).

Step 2:

- ➔ If n (number of values) is odd : median is the middle value:
  - ◆ Position (n+1)/2.
- ➔ If n is even → median is the average of the two middle values:
  - ◆ Average of values at positions: n/2 and (n/2)+1.

Univariate analysis

Median

The median is the middle value of a dataset when the values are arranged in order (from smallest to largest).

Example 1 (odd number of values):

Data: [10, 14, 18, 22, 26]
→ 5 values → middle = 3rd value
→ Median = 18

Example 2 (even number of values):

Data: [10, 14, 18, 22, 26, 30]
→ 6 values → middle = average of 3rd and 4th values
→ (18 + 22) / 2 = 20

Univariate analysis

Mode

The mode is the value (or class) that appears most frequently in a dataset.
It represents the most common or typical observation.

➔ Simply find the value with the highest frequency.

Useful for:

Works with all data types: numerical, categorical, or ordinal.
Indicates the most common value or category.
Helpful for nominal data (e.g., favorite color, best seller, product choice).
In skewed distributions, mode often differs from mean and median.

Univariate analysis

Mode

The mode is the value (or class) that appears most frequently in a dataset.
It represents the most common or typical observation.

Example:

Data = [4, 5, 6, 6, 7, 8, 8, 8, 9]
→ Mode = 8, because it appears most often.

If two values tie for highest frequency:
➔ The data is bimodal (two modes).
If more than two:
➔ The data is multimodal.
If all values occur once:
➔ There is no mode.

Univariate analysis

Relationship Between Mean, Median, and Mode

This relationship shows how the mean, median, and mode are related to each other in a moderately skewed distribution (not perfectly symmetrical but not extremely skewed either).

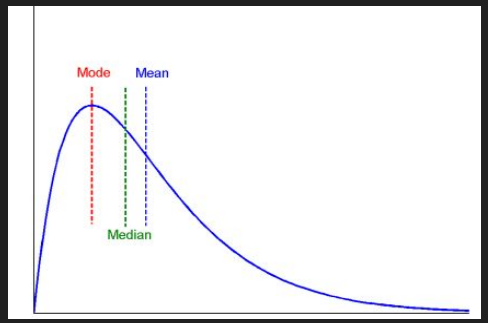
$$mean - mode = 3 \times (mean - median)$$

It helps estimate one measure (usually the mode) if the other two are known.

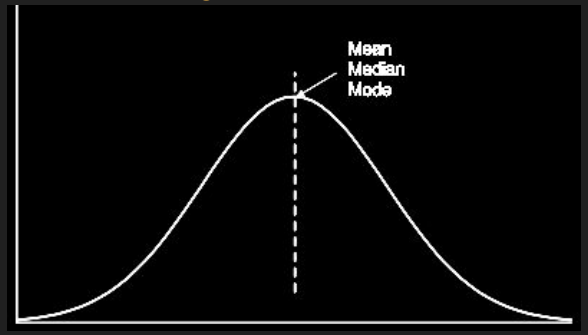
Type of Distribution	Relationship	Shape
Symmetrical	Mean = Median = Mode	Bell-shaped
Positively Skewed (Right-skewed)	Mean > Median > Mode	Tail on the right
Negatively Skewed (Left-skewed)	Mode > Median > Mean	Tail on the left

Univariate analysis

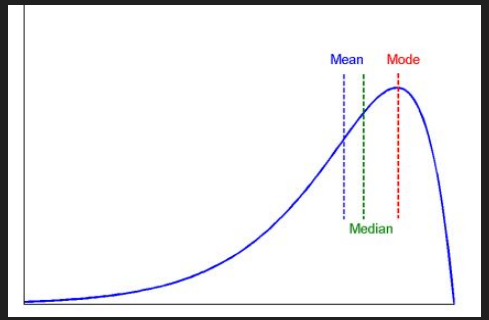
Positively Skewed



Symmetrical



Negatively Skewed



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Univariate analysis

Autres mesures

Midrange (milieu de gamme)
The midrange is the average of the smallest and largest values in a dataset.
It represents the midpoint of the data's range.

Midrange = (min + max) / 2

Range
The range is a simple measure of dispersion (spread) in a dataset.
It tells you how far apart the smallest and largest values are.

Range = max - min

Univariate analysis

Measuring Data Dispersion

Dispersion describes how spread out or scattered the data values are around the center (mean or median).

Measuring dispersion helps to:

- ➔ Understand data variability (consistency vs instability).
- ➔ Compare two or more datasets.
- ➔ Identify outliers or data spread.
- ➔ Interpret central tendency correctly (mean alone can be misleading).

Univariate analysis

Measuring Data Dispersion

Quartile Q1	The first quartile (Q1) is the (n/4)th value of the ordered dataset. (25% of the data are below Q1).
Quartile Q3	The first quartile (Q3) is the (3 . n/4)th value of the ordered dataset. (75% of the data are below Q3).
5 Numbers summary	(MIN, Q1, Q2, Q3, MAX)
Interquartile Range (IQR)	IQR = Q3 - Q1
Outliers **(Valeurs aberrantes)**	Values > Q3 + 1.5 * IQR Values < Q1 - 1.5 * IQR
Variance	$\sigma^2 = \frac{\sum(x_i - \bar{x})^2}{N}$
Standard deviation **(Ecart-type)**	$\sigma = \sqrt{\text{Variance}}$

Univariate analysis

Measuring Data Dispersion

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Variance	$\sigma^2 = \frac{\sum(x_i - \bar{x})^2}{N}$
Standard deviation **(Ecart-type)**	$\sigma = \sqrt{\text{Variance}}$

What if the Q1 position is odd ??

Univariate analysis

Missing Value / Data

A missing value is simply a data point that's not recorded or unavailable in a dataset. It can be represented by:

NaN, NULL, None, (blank), ?, "", or 0

Causes of Missing Values:

- ➔ Data entry errors or omissions
- ➔ Sensor or system failures
- ➔ Survey questions left unanswered
- ➔ Data not collected for certain cases

Univariate analysis

Unique Values
A unique value is a value that appears only once in a column or dataset.

Why checking unique values:

- ➔ Detect duplicates: Check if all values in an ID column are unique
- ➔ Understand variability: Columns with few unique values may bring low information
- ➔ Identify categorical features: Few unique values means useful for encoding categories (gender, region, status)
- ➔ Spot data quality issues: Unexpected duplicates signal errors

Univariate analysis

Outliers

An outlier is a data point that is very different from the rest of the data. It lies far away from most other values.

Example:

Data: [10, 12, 11, 13, 12, 200]
Here, 200 is an outlier : it is far from the other values (around 10–13).

Causes of Outliers::

- ➔ Data entry errors (e.g., typing 200 instead of 20)
- ➔ Measurement or sensor error
- ➔ Natural extreme cases (e.g., very high income)

Univariate analysis

Outliers

An outlier is a data point that is very different from the rest of the data. It lies far away from most other values.

Method	Description	Rule
Z-Score Method	Measures how many standard deviations a value is from the mean	Outlier if $Z > 3$ or $Z < -3$
IQR Method	Based on quartiles (Q1, Q3)	Outlier if value $< Q1 - 1.5 \times IQR$ or value $> Q3 + 1.5 \times IQR$
Boxplot	Visual tool showing outliers as points beyond whiskers	Visual analysis
Scatter Plot	Helps visualize isolated data points	Visual analysis

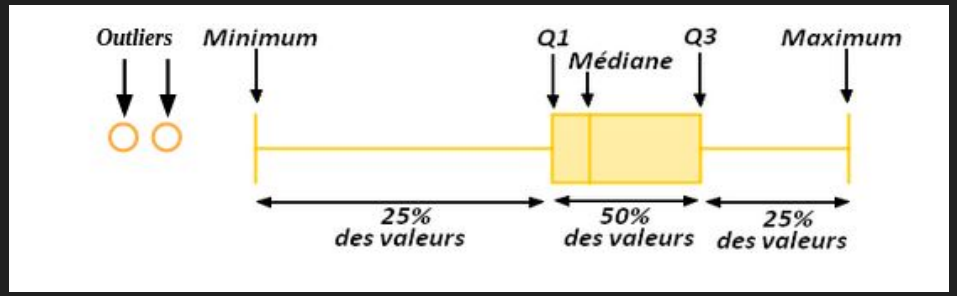
Univariate analysis

Visual Analysis

Boxplots, Histograms, Bar Charts, and Quantile Plots are four fundamental data visualization tools used in exploratory data analysis.

BoxPlots (Boite à moustache)

Shows the distribution of data, spread, and outliers using quartiles.



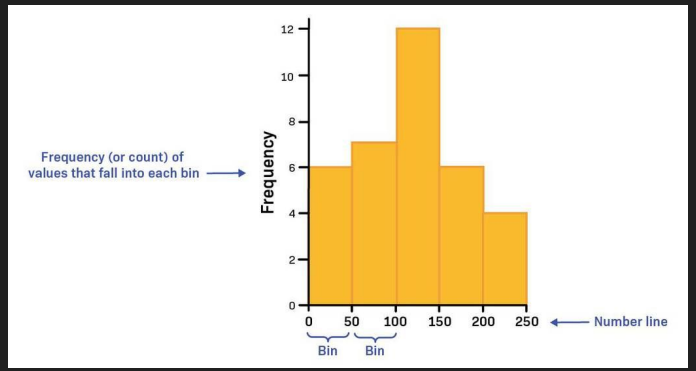
- ➔ Detects outliers
- ➔ Shows symmetry or skewness
- ➔ Compares distributions across groups

Univariate analysis

Histograms

Shows the frequency distribution of numerical (continuous) data.

- ➔ Divides data into bins (intervals)
- ➔ The height of each bar = number (or %) of values in that range



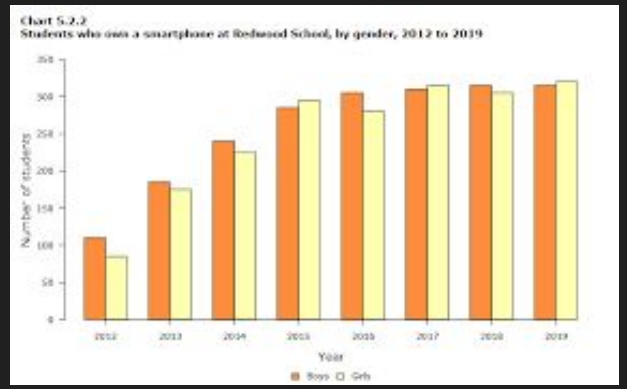
- ➔ Reveals shape of distribution (normal, skewed, bimodal, etc.)
- ➔ Detects gaps or clusters
- ➔ Helps identify outliers indirectly

Univariate analysis

Bar Chart

Displays frequencies or proportions of categorical data.

- ➔ Each bar represents a category
- ➔ Bar height = count or percentage

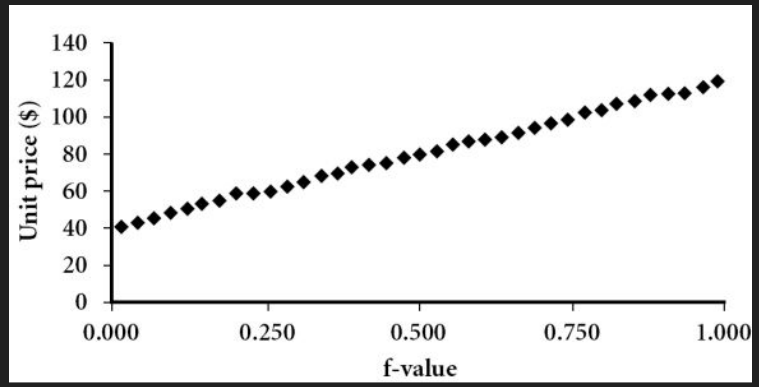


- ➔ Compares categories (modal, bimodal,..)
- ➔ Highlights most common or least common groups

Univariate analysis

Quantile Plot

Allows evaluation of both the overall behavior and unusual occurrences. Represents the ordered values X_i as a function of their relative frequencies f_i(%) , such that: $f_i = (i - 0.5) / n$ (i is the rank (or position) of the value X_i)



- ➔ Detects skewness or non-normality
- ➔ Identifies outliers
- ➔ Evaluates distribution fit

Bivariate analysis

Definition

Once the univariate analysis is complete, the next step is to perform **bivariate analysis**, which involves analyzing the relationship between two variables. This helps to understand how the variables are related and identify any patterns or correlations in the data.

Type of Variables	Techniques Used
Numerical – Numerical	Scatter plot, correlation coefficient, Q-Q plot
Categorical – Numerical	Box plot, group mean comparison (ANOVA, t-test)
Categorical – Categorical	Chi-square test

Bivariate analysis

Numerical – Numerical

Scatter Plot

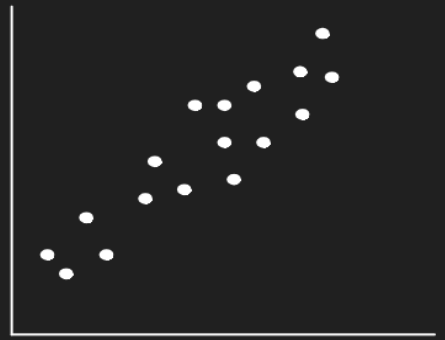
A scatter plot is a graph in which the values of two variables are plotted along two axes, with each individual data point represented by a dot on the graph. The pattern formed by the dots can reveal the nature of the relationship between the variables.

Bivariate analysis

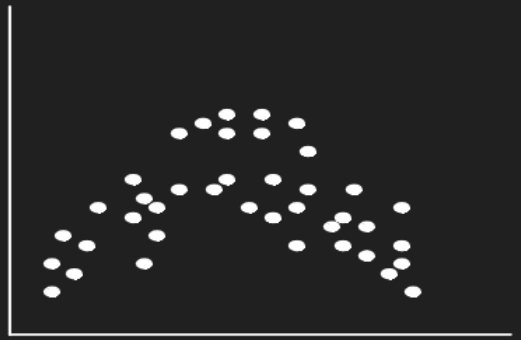
Numerical – Numerical

Scatter Plot

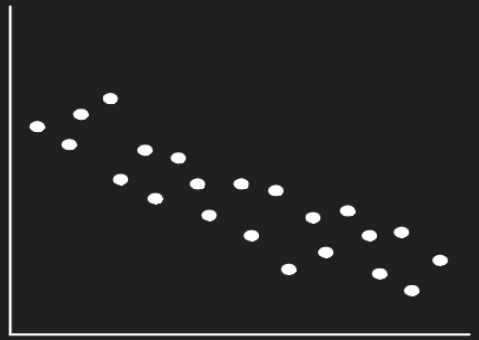
The pattern formed by the dots can reveal the nature of the relationship between the variables.



Positively correlated



L half PC & R half NC



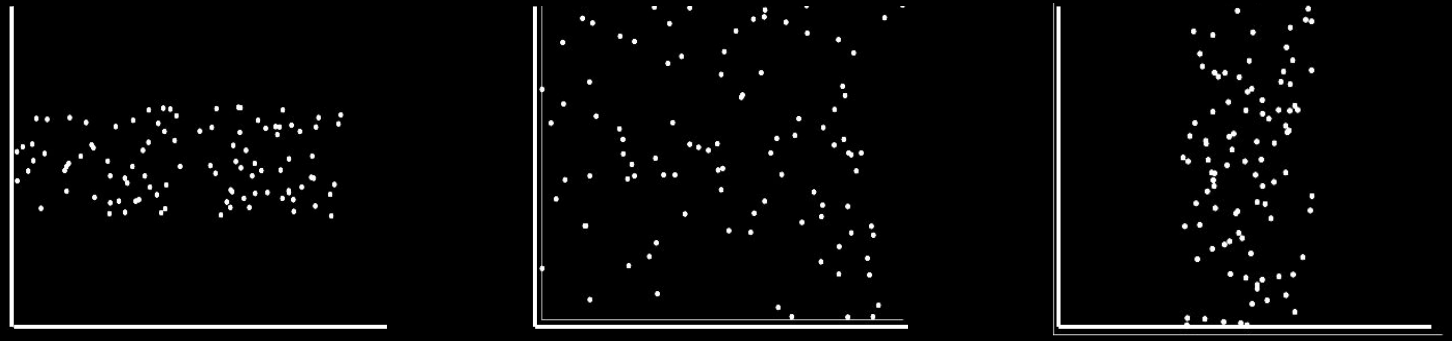
Negatively correlated

Bivariate analysis

Numerical – Numerical

Scatter Plot

The pattern formed by the dots can reveal the nature of the relationship between the variables.



Uncorrelated

Bivariate analysis

Numerical – Numerical

Correlation

Correlation is a measure of the strength and direction of the relationship between two variables. It is used to identify the extent to which two variables are related to each other.

Correlation is usually measured using a correlation coefficient, which ranges from -1 to +1. A coefficient of +1 indicates a perfect positive correlation, a coefficient of -1 indicates a perfect negative correlation, and a coefficient of 0 indicates no correlation.

Bivariate analysis

Numerical – Numerical

Correlation

Correlation is a measure of the strength and direction of the relationship between two variables. It is used to identify the extent to which two variables are related to each other.

Two main types: Pearson correlation and Spearman correlation.
Pearson correlation: is used when both variables are normally distributed,
Spearman correlation: is used when one or both variables are not normally distributed.

Bivariate analysis

Numerical – Numerical

Correlation

Pearson correlation: is used when both variables are normally distributed

$$r_{A,B} = \frac{\sum_{i=1}^{n}(a_i - \bar{A})(b_i - \bar{B})}{(n-1)\sigma_A\sigma_B} = \frac{\sum_{i=1}^{n}(a_i b_i) - n\bar{A}\bar{B}}{(n-1)\sigma_A\sigma_B}$$

where n is the number of tuples, $\bar{A}$ and $\bar{B}$ are the respective means of A and B, σA and σB are the respective standard deviation of A and B, and $\Sigma(a_i b_i)$ is the sum of the AB cross-product.

Bivariate analysis

Numerical – Numerical

Correlation

Spearman correlation: assesses monotonic relationships whether linear or not. It is most used when the variable are not normally distributed.

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

d_i = difference between the ranks of A_i and B_i
n = number of observations

Bivariate analysis

Numerical – Numerical

Correlation

Spearman correlation: assesses monotonic relationships whether linear or not. It is most used when the variable are not normally distributed.

When Spearman is better than Pearson:

- When data are not normally distributed.
- When variables have ordinal values (ranks, scores).
- When the relationship is monotonic but not linear (curves that always increase/decrease but not at a constant rate).
- When outliers might distort Pearson correlation.

Bivariate analysis

Numerical – Numerical

Quantile–Quantile Plot (Q-Q Plot)

Graphs the quantiles of one univariate distribution against the corresponding quantiles of another. If the plot is close to the f(x)=x line it means the 2 distributions are identical.

Use cases:

- Show if there is a shift in going from one distribution to another.
- Compare 2 datasets or 2 attributes of same nature (Prices of brand A Vs prices of brand B)
- Validating data transformations (log, normalisation, ...).

Bivariate analysis

Numerical – Numerical

Quantile–Quantile Plot (Q-Q Plot)

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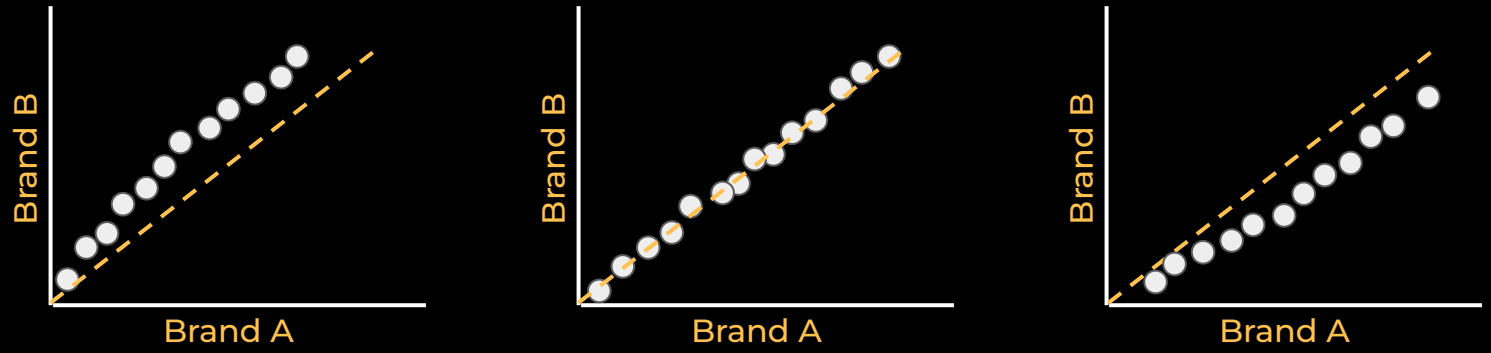


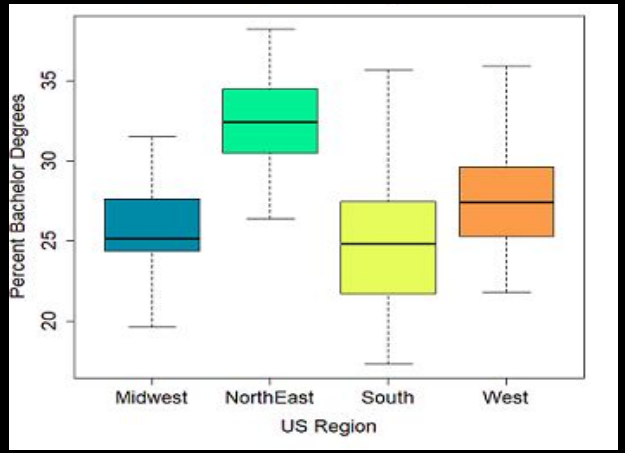
Bivariate analysis

Categorical – Numerical

Boxplots

In bivariate analysis, a boxplot is a powerful visualization tool used when you have: one categorical variable and one numerical variable.

It helps you compare the distribution of the numerical variable across the categories of the categorical variable.



Bivariate analysis

Categorical – Numerical

Group mean comparison (T-test)
Group mean comparison tests are used to determine whether the average (mean) of a numerical variable differs significantly between two or more groups of a categorical variable.

Find a step by step example here:
https://www.statisticshowto.com/probability-and-statistics/t-test/

Bivariate analysis

Categorical – Categorical

Chi-Square : X^2

The Chi-square test is a statistical test used to determine whether there is a significant relationship between two categorical variables.
It compares the observed frequencies (what you actually see in your data) with the expected frequencies (what you would expect if there were no relationship).

$$\chi^2 = \sum \frac{(Observed - Expected)^2}{Expected}$$

O_i = observed frequency
E_i = expected frequency
The sum is taken over all possible categories or cells in a contingency table.

Bivariate analysis

Categorical – Categorical
Chi-Square : X^2

Income Level

Tax Bill	Low	Medium	High	
For	213 (209.9)	203 (187.2)	182 (200.9)	598
Against	138 (141.1)	110 (125.8)	154 (135.1)	402
	351	313	336	1000

$$E_{ij} = \frac{(Row\ Total) \times (Column\ Total)}{Grand\ Total}$$

df = (number of rows - 1) × (number of columns - 1)

H_0 : voter's opinion concerning the new tax bill and their income level is independent

H_1 : voter's opinion concerning the new tax bill and their income level is not independent

Test Statistic: $\chi^2 = \sum \frac{(Observed - Expected)^2}{Expected} = \sum \frac{(O-E)^2}{E}$, where $df = (2-1)(3-1) = 2$

$$\chi^2 = \frac{(213-209.9)^2}{209.9} + \frac{(203-187.2)^2}{187.2} + \frac{(182-200.9)^2}{200.9} + \frac{(138-141.1)^2}{141.1} + \frac{(110-125.8)^2}{125.8} + \frac{(154-135.1)^2}{135.1} = 7.85$$

Critical Value: $\chi^2_a = \chi^2_{(0.05,2)} = 5.99$

Sine $\chi^2 = 7.85 > \chi^2_a = 5.99$ we reject the null hypothesis and conclude that voter opinion concerning the new tax reform bill and income level are not independent.

Multivariate analysis

Definition

Multivariate analysis is a set of statistical techniques used to analyze data that involves more than two variables simultaneously.
It helps identify relationships, patterns, and dependencies among multiple variables.

Some common techniques used in multivariate analysis include:

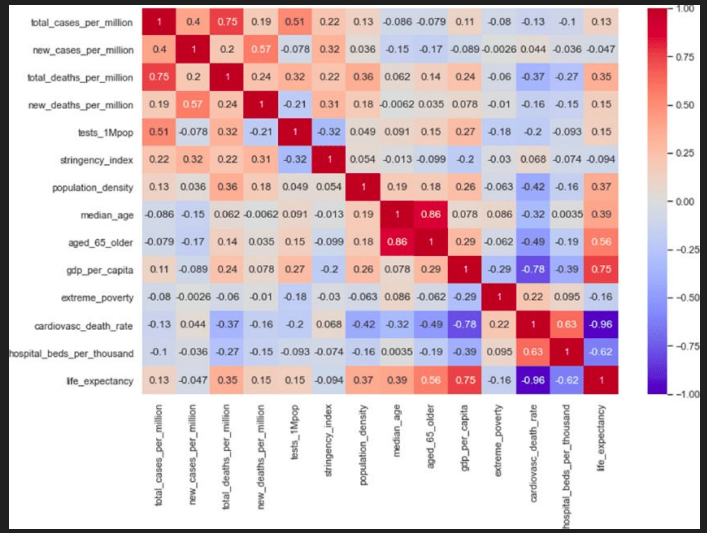
- Heatmaps
- Correlation matrices
- PCA visualization
- Clustering visualization

Multivariate analysis

Heatmaps / Correlation Matrix

A heatmap is a representation of data that uses a color-coding system to represent different values of correlation.

It is used to display the correlation between multiple variables, making it easier to identify patterns and trends in large datasets.



Multivariate analysis

Principal Component Analysis (PCA)

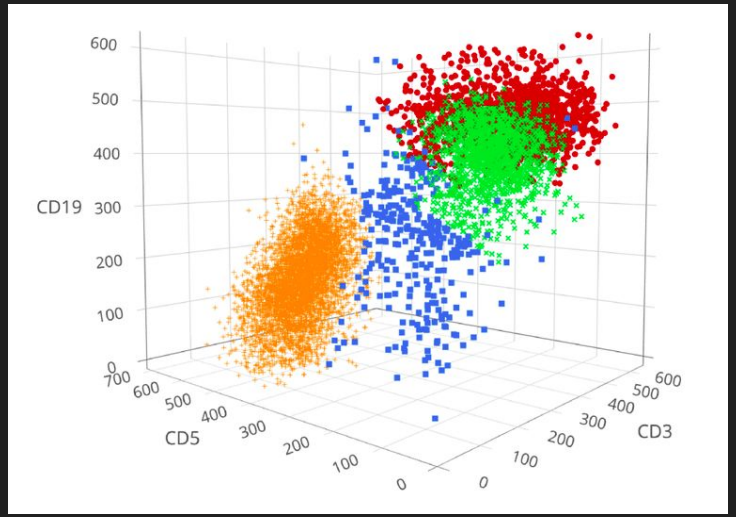
PCA is a dimensionality reduction technique that transforms a large set of correlated variables into a smaller set of uncorrelated components (called principal components), while preserving as much of the data's variability as possible.

PCA projects high-dimensional data into 2D or 3D, allowing the visualization of the relationships and patterns that would otherwise be hidden.

Other methods: t-SNE, UMAP, ...etc.

Multivariate analysis

Cluster Analysis

Clustering, is used to group a set of objects in a way that objects in the same group (cluster) are more similar to each other than to those in other groups.

The objective of cluster analysis is to find a structure or pattern in the data that can be useful in discovering relationships or identifying groups within the data.